

Introduction to Integers

Jonathan R. Bacolod

The set of **integers** consists of the **negative whole numbers**, **zero**, and the **positive whole numbers**.

Symbol for Integers

The symbol for the set of integers is $\mathbb{Z}$.

Whenever you see $\mathbb{Z}$ in math, think “integers.”

The Set of Integers

The set of integers can be written as

$$\mathbf{Z} = \{\dots, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots\}.$$

$$\dots \quad -3 \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \quad \dots$$

The dots mean the pattern continues forever
in both directions.

What Integers Are NOT

Integers do **not** include:

- ▶ Fractions (like $\frac{1}{2}$)
- ▶ Decimals (like 0.75)
- ▶ Irrational numbers (like $\sqrt{2}$)

Whole Numbers vs. Integers

Every **whole number** is an integer, but not every integer is a whole number, because integers also include negative numbers.

Number	Whole Number?	Integer?
5	Yes	Yes
0	Yes	Yes
-5	No	Yes

Examples

Example 1

Is -9 an integer?

Yes. -9 is a negative whole number, so it belongs to $\mathbf{Z}$.

Example 2

Is $\frac{3}{4}$ an integer?

No. $\frac{3}{4}$ is a fraction, and integers do not include fractions.

Example 3

Is 6.0 an integer?

Yes. Even though it is written with a decimal point, 6.0 is the same value as the whole number 6.

Example 4

Is -2.5 an integer?

No. -2.5 is a decimal that falls between -3 and -2 , so it is not an integer.

Practice Exercises

Practice

Identify whether each number is an integer or not.

Practice 1

15

Practice 2

— 23

Practice 3

0

Practice 4

7

—

2

Practice 5

5.8

Practice 6

— 11

Practice 7

—0.75

Practice 8

100

Activities

Activity

Identify whether each number is an integer or not.

Activity 1

42

Activity 2

— 17

Activity 3

0

Activity 4

9

|

5

Activity 5

8.25

Activity 6

—30

Activity 7

—1.4

Activity 8

256

Thank you!